Life Cycles of

Stars

What is a star?

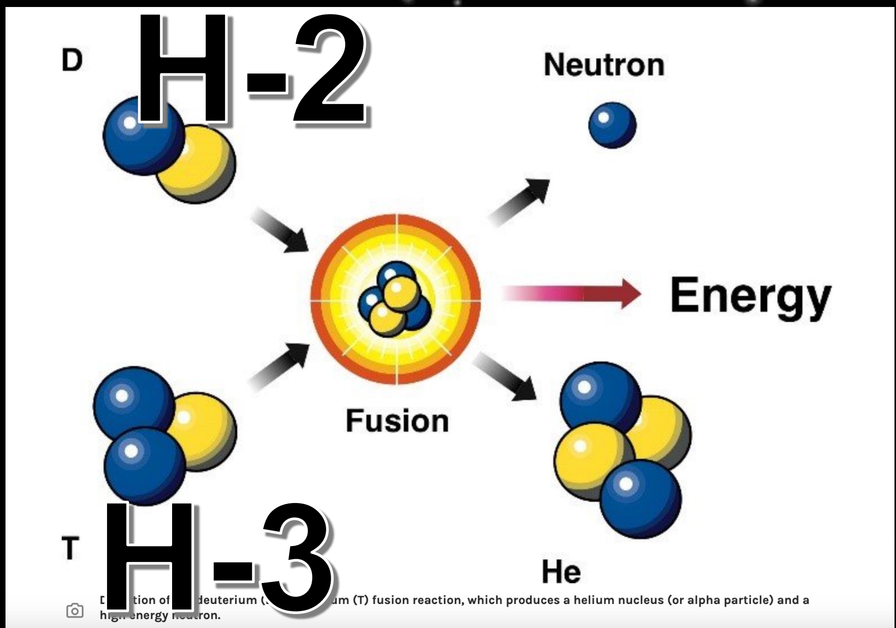
- Stars are giant balls of hot gas – mostly hydrogen, with some helium and small amounts of other elements.
- The gas is so hot that it ionizes (loses electrons) and becomes plasma.
- Every star has its own life cycle, ranging from a few million to trillions of years, and its properties change as it ages.

Do Stars Burn?

A star is not like a fire on Earth.

Stars shine because of **nuclear fusion**.

Energy from the nuclear reactions is released as **electromagnetic radiation**



This reaction releases energy because the mass of the helium atom is less than the combined mass of both Hydrogen atoms. The leftover mass is released as energy according to the equation $E=mc^2$.

Measuring Stars

DISTANCE: Stars are <u>really</u> far apart!

– Measured in **light-years**

- The distance which a particle of light travels through a vacuum in one year
- Exactly 9,460,730,472,580.8 km (9.5 trillion)

- Speed of Light = 299,792 km per second

Astronomical Units (AU):

- Average distance between Earth and the Sun
- 1 AU = 149,597,870.7 km. (150 million km)

Measuring Stars

- **Magnitude** (brightness)
  - Smaller values represent brighter objects than larger values
  - **Apparent magnitude**
    - How bright a star appears to be from Earth
  - **Absolute magnitude (luminosity)**
    - How bright a star actually is

Why would a star's apparent magnitude be different from the absolute magnitude?

Measuring Stars

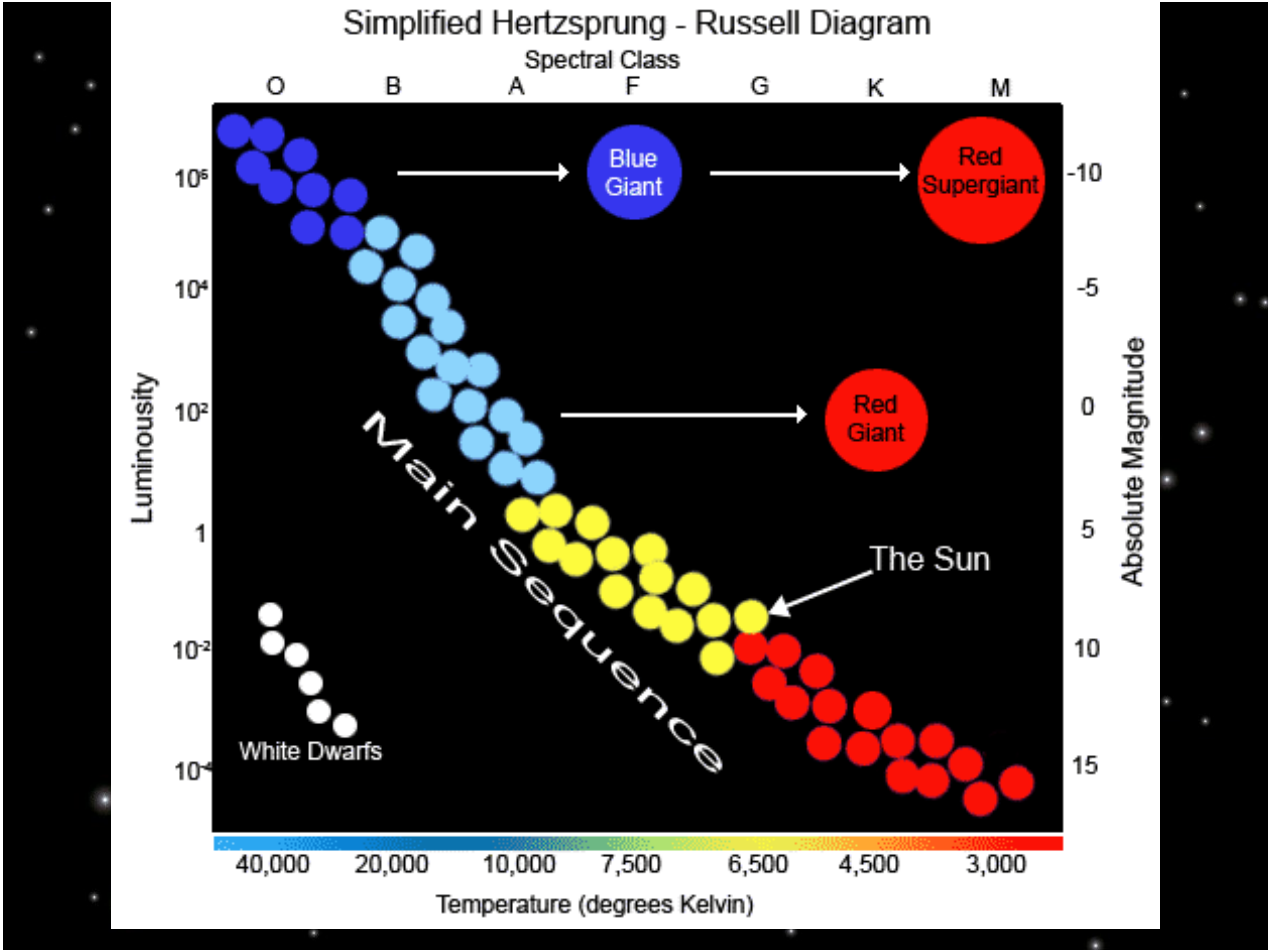
Temperature & Color

The color of a star indicates the temperature of the star. Brighter stars tend to be hotter.

Decreasing temperature (bright to dim):

O, B, A, F, G, K, M

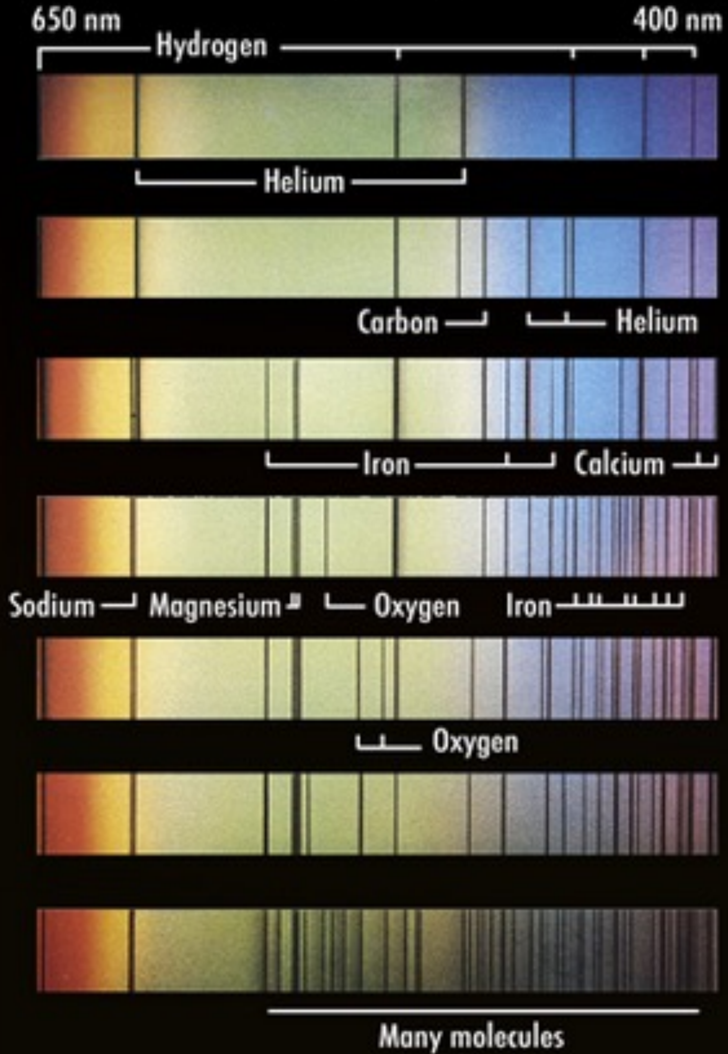
[**O**ld **B**ob **A**lways **F**avors **G**reen **K**etchup **M**ore]



Simplified Hertzsprung - Russell Diagram
Spectral Class
O
B
A
F
G
K
M
Blue Giant
Red Supergiant
Red Giant
Main Sequence
The Sun
White Dwarfs
Luminousity
10^6
10^4
10^2
1
10^{-2}
10^{-4}
Absolute Magnitude
-10
-5
0
5
10
15
40,000
20,000
10,000
7,500
6,500
4,500
3,000
Temperature (degrees Kelvin)

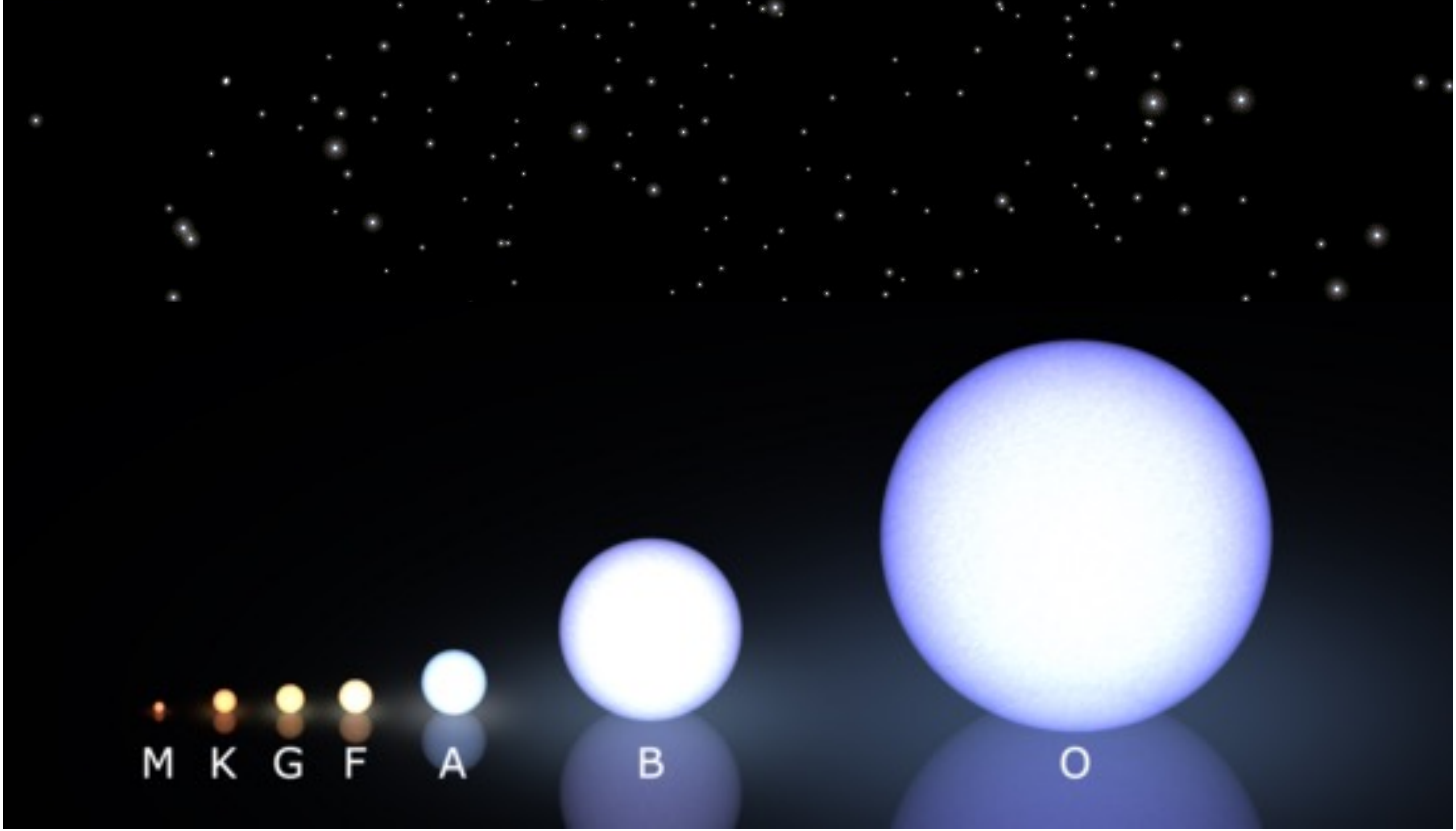
STELLAR CLASSIFICATION (MAIN-SEQUENCE)

SPECTRUM



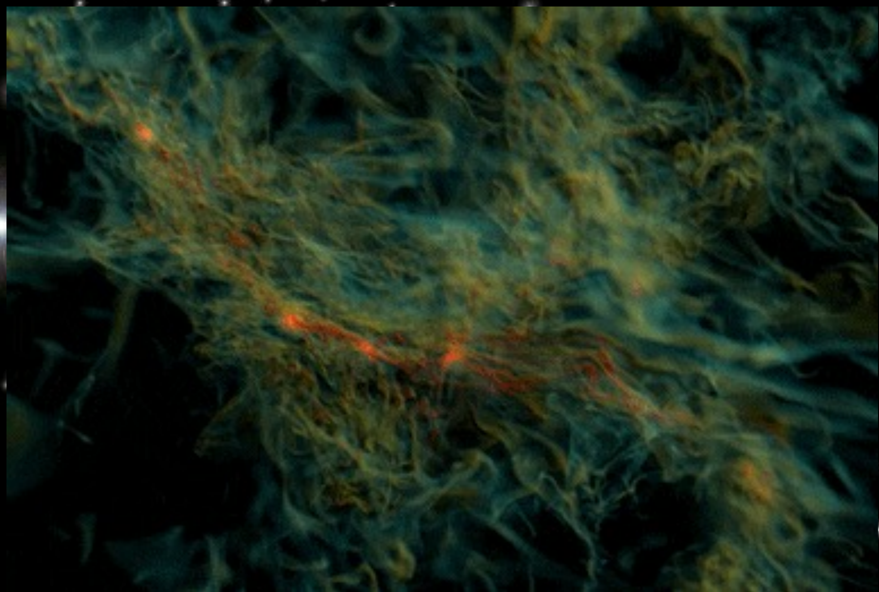
CLASS	COLOR	TEMP. (C°)	HYDROGEN	SIZE(SOLAR R)	% OF ALL
O		>30,000	WEAK	>6.6	~0.00003%
B		9,700 - 30,000	MEDIUM	1.8 - 6.6	0.13%
A		7,200 - 9,700	STRONG	1.4 - 1.8	0.6%
F		5,700 - 7,200	MEDIUM	1.1 - 1.4	3%
G		4,900 - 5,700	WEAK	.9 - 1.1	7.6%
K		3,400 - 4,900	VERY WEAK	.7 - .9	12.1%
M		2,100 - 3,400	VERY WEAK	<.7	76.5%

Size Comparison of Main Sequence Stars



Life Cycle of Stars

- Begin as a cloud of dust and gas called a **nebula**
- Gravity may cause the nebula to contract
- Matter in the gas cloud will begin to condense into a dense region called a <u>protostar</u>
- The protostar continues to condense, it heats up. Eventually, it reaches a **critical mass** and nuclear fusion begins. This is the <u>main sequence</u> phase




HBC 1 is a newly formed star.
It is not yet in its main sequence.

Main Sequence Stars

- **The Main sequence** is the longest part of a star's life cycle.
- They fuse hydrogen isotopes into helium in the core, which happens because of the immense pressure there.
- About 90% of the stars in the universe are **main sequence stars**

Life Cycle of Stars

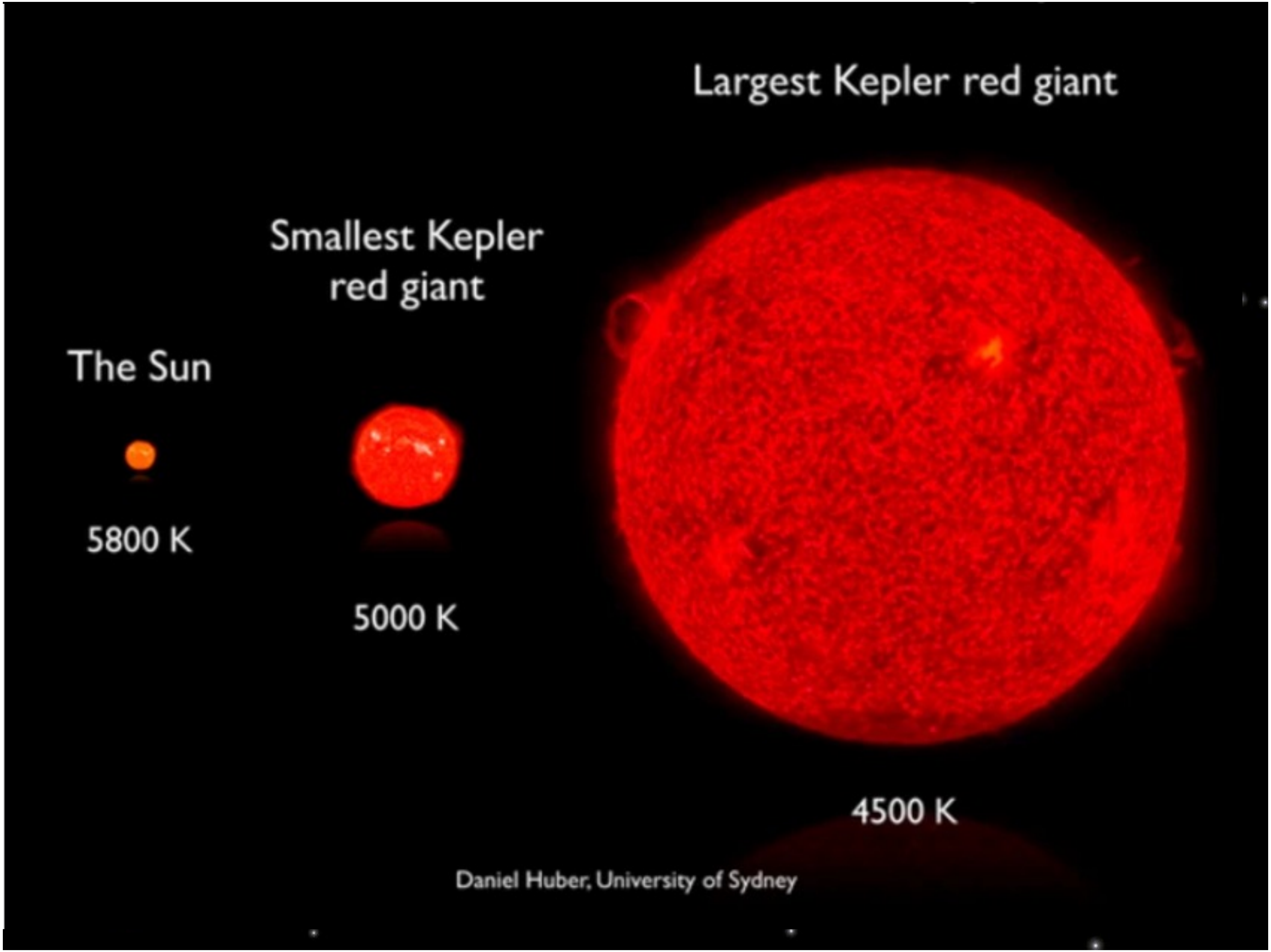
Life span of a star depends on its size.

- Very large, massive stars burn their fuel much faster than smaller stars
- Their main sequence may last only a few hundred thousand years
- Smaller stars will live on for billions of years because they burn their fuel much more slowly

- Eventually, the star’s hydrogen fuel will begin to run out.

Life Cycle of Stars

- When the star's core is out of Hydrogen, it will try to fuse other elements.
- This creates the heavier elements on the Periodic Table.
- It will expand into a <u>red giant</u>
- Massive stars will become <u>red supergiants</u>
- This phase will last until the star exhausts its remaining fuel
- At this point the star will collapse


Largest Kepler red giant
Smallest Kepler
red giant
The Sun
5800 K
5000 K
4500 K
Daniel Huber, University of Sydney

Life Cycle of Stars

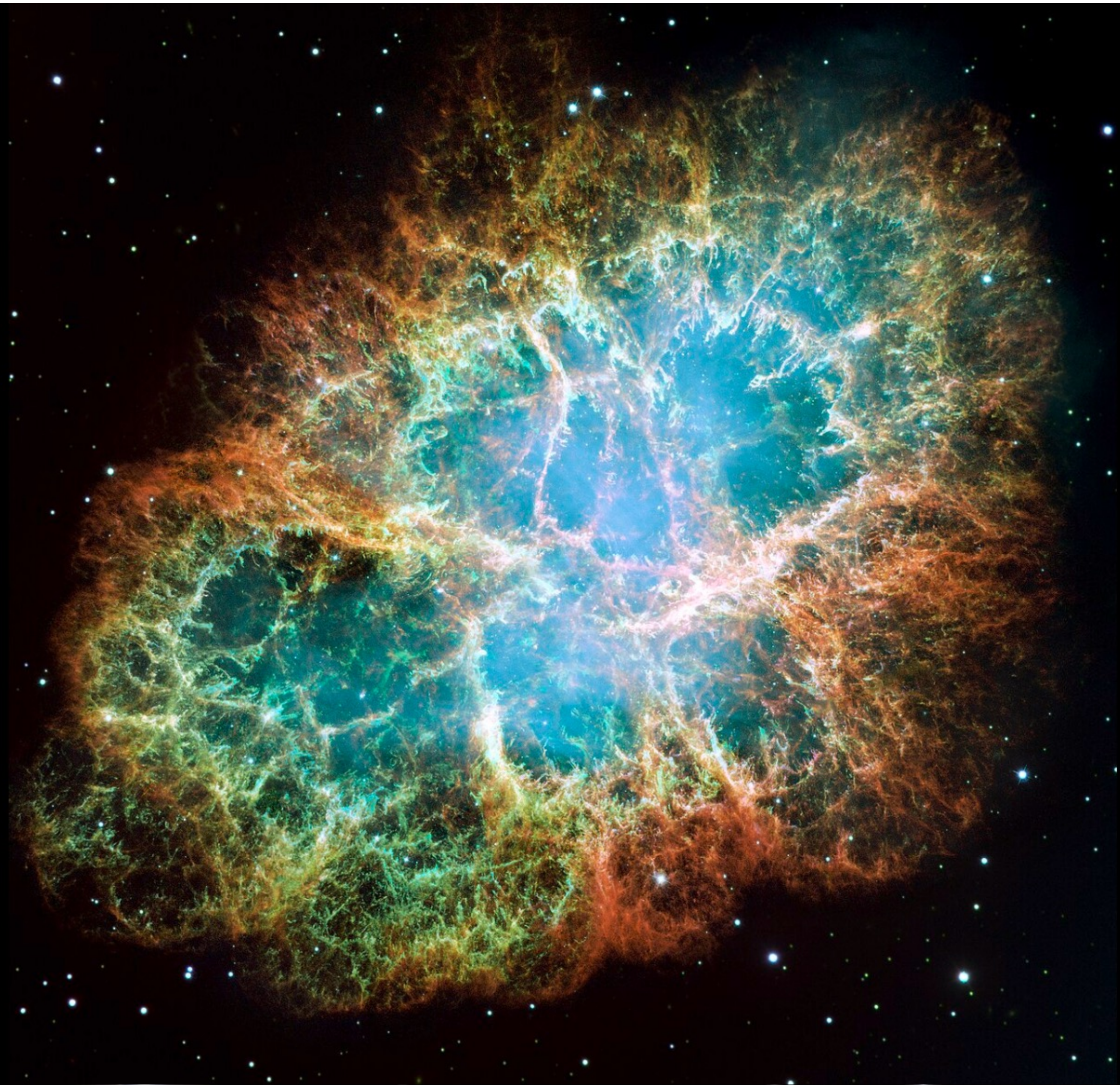
- Most average stars will blow away their outer atmospheres to form a planetary nebula
- Their cores will remain behind and burn as a white dwarf until they cool down
- What will be left is a dark ball of matter known as a black dwarf

The Southern Ring Nebula is a planetary nebula about 2,500 light-years away.

Life Cycle of Stars

- If the star is massive enough, the collapse will trigger a violent explosion known as a **supernova**
- If the remaining mass of the star is about 1.4 times that of our Sun, the core is unable to support itself and it will collapse further to become a **neutron star**

The fastest-spinning neutron star known is PSR J1748-2446ad, rotating at a rate of 716 times per second



The Crab Nebula was formed by a supernova recorded in 1054. At its center is a neutron star.

Life Cycle of Stars

- The Neutron star's matter will be compressed so tightly that its atoms are compacted into a dense shell of neutrons. If the remaining mass of the star is more than about three times that of the Sun, it will collapse completely.
- What is left behind is an intense region of gravity called a black hole

Gravitational Lensing happens when a black hole's gravity bends the light around it. This is the main way astronomers detect black holes.

Life Cycle of Stars